

1

Problem

- LLMs know everyday procedures but have not experienced the physical consequences of their words.
- A plausible text plan may reference unavailable objects or actions the robot cannot execute.
- The goal is zero-shot execution of abstract, ambiguous and long-horizon language instructions.

2

Core idea

TASK x WORLD GROUNDING

- The LLM scores whether each skill description is a useful next step for the instruction.
- A value function estimates whether that skill can succeed from the current physical state.
- Their product favors actions that are simultaneously contextually appropriate and feasible.

3

Planning loop

SAYCAN ALGORITHM

- Enumerate the robot's admissible low-level skills and score every candidate.
- Execute the best skill, observe the new state, append the skill to dialogue history and re-plan.
- Repeat until the model selects the termination token 'done'.

SAYCAN

Do As I Can, Not As I Say

Grounding Language in Robotic Affordances | CoRL 2022

Combine an LLM's semantic plan score with a robot skill's learned affordance, selecting actions that are both useful and physically feasible.

LLM planningaffordancerobotics

4

Mathematics

PROBABILISTIC FUSION

- $s_{t^*} = \operatorname{argmax}_s p_{\text{LLM}}(l_s \mid \text{instruction}, \text{history}) \times V_s(\text{state}_t)$
- p_{LLM} provides task-grounding; V_s approximates $P(\text{skill succeeds} \mid \text{state})$ and provides world-grounding.
- Skill policies come from multi-task behavior cloning or RL; TD-trained value functions encode affordances.

5

Evidence

REAL ROBOT RESULTS

- 101 instructions, 7 families, 15 objects and 5 locations on a mobile manipulator.
- Mock kitchen: 84% plan success and 74% execution; real kitchen: 81% and 60%.
- Without affordances, plan success falls to 67%; PaLM-SayCan also beats FLAN-SayCan (84/74 vs 70/61).

6

Meaning & limits

TAKEAWAY

- Value functions become the LLM's 'hands and eyes'; the LLM supplies high-level semantic knowledge.
- New skills can be added modularly, and stronger LLMs directly improve robot planning.
- The skill library is the main bottleneck; LLM bias, execution failures and compounding errors remain.